Human cancer cell antiproliferative and antioxidant activities of Juglans regia L.

Several studies suggest that regular consumption of nuts, mostly walnuts, may have beneficial effects against oxidative stress mediated diseases such as cardiovascular disease and cancer. Walnuts contain several phenolic compounds which are thought to contribute to their biological properties. The present study reports the total phenolic contents and antioxidant properties of methanolic and petroleum ether extracts obtained from walnut (Juglans regia L.) seed, green husk and leaf. The total phenolic contents were determined by the Folin-Ciocalteu method and the antioxidant activities assessed by the ability to quench the stable free radical 2,2'-diphenyl-1-picrylhydrazyl (DPPH) and to inhibit the 2,2'-azobis(2-amidinopropane) dihydrochloride (AAPH)-induced oxidative hemolysis of human erythrocytes. Methanolic seed extract presented the highest total phenolic content (116 mg GAE/g of extract) and DPPH scavenging activity (EC(50) of 0.143 mg/mL), followed by leaf and green husk. In petroleum ether extracts, antioxidant action was much lower or absent. Under the oxidative action of AAPH, all methanolic extracts significantly protected the erythrocyte membrane from hemolysis in a time- and concentration-dependent manner, although leaf extract inhibitory efficiency was much stronger (IC(50) of 0.060 mg/mL) than that observed for green husks and seeds (IC(50) of 0.127 and 0.121 mg/mL, respectively). Walnut methanolic extracts were also assayed for their antiproliferative effectiveness using human renal cancer cell lines A-498 and 769-P and the colon cancer cell line Caco-2. All extracts showed concentration-dependent growth inhibition toward human kidney and colon cancer cells. Concerning A-498 renal cancer cells, all extracts exhibited similar growth inhibition activity (IC(50) values between 0.226 and 0.291 mg/mL), while for both 769-P renal and Caco-2 colon cancer cells, walnut leaf extract showed a higher antiproliferative efficiency (IC(50) values of 0.352 and 0.229 mg/mL, respectively) than green husk or seed extracts. The results obtained herein strongly indicate that walnut tree constitute an excellent source of effective natural antioxidants and chemopreventive agents.